量子力学

内积：$\langle\psi|\varphi\rangle=\int\psi^*(\vec r)\varphi(\vec r)d\vec r$; $\langle\psi|\varphi\rangle=\langle\varphi|\psi\rangle^*$; $\langle\psi|\sum_i c_i\varphi_i\rangle=\sum_i c_i\langle\psi|\varphi_i\rangle$

算符：$\hat x:\ \psi(x)\mapsto x\psi(x)$ $\quad\tilde\psi(p)\mapsto i\hbar\frac{\partial\tilde\psi(p)}{\partial p}$ $\quad\langle\sum_i c_i\psi_i|\varphi\rangle=\sum_i c_i^*\langle\psi_i|\varphi\rangle$

$\hat p:\ \psi(x)\mapsto -i\hbar\frac{\partial\psi(x)}{\partial x}$ $\quad\tilde\psi(p)\mapsto p\tilde\psi(p)$ $\quad[\sum_i a_i\hat A_i,\sum_j b_j\hat B_j]=\sum_i\sum_j a_ib_j[\hat A_i,\hat B_j]$

对易子：$[\hat A,\hat B]=\hat A\hat B-\hat B\hat A$; $[\hat A,\hat B\hat C]=[\hat A,\hat B]\hat C+\hat B[\hat A,\hat C]$; $[\hat A\hat B,\hat C]=\hat A[\hat B,\hat C]+[\hat A,\hat C]\hat B$

正则对易关系：$[\hat x,\hat p]=i\hbar$; $[\hat x,f(\hat p)]=i\hbar\left(\frac{\partial f(p)}{\partial p}\right)_{p\to\hat p}$; $[f(\hat x),\hat p]=i\hbar\left(\frac{\partial f(x)}{\partial x}\right)_{x\to\hat x}$

厄密共厄：$\langle\hat O^\dagger\psi|\varphi\rangle=\langle\psi|\hat O\varphi\rangle$; $(\hat A\hat B)^\dagger=\hat B^\dagger\hat A^\dagger$; $(\sum_i c_i\hat O_i)^\dagger=\sum_i c_i^*\hat O_i^\dagger$

厄密算符的本征基：正交归一：$\langle\psi_{\hat O=\lambda}|\psi_{\hat O=\lambda'}\rangle=\begin{cases}\delta_{\lambda\lambda'} & \text{离散}\\ \delta(\lambda-\lambda') & \text{连续}\end{cases}$ $\quad\hat O|\psi_{\hat O=\lambda}\rangle=\lambda|\psi_{\hat O=\lambda}\rangle$

恒等算符分解：$\hat 1=\text{“}\sum_{\lambda,k}\text{”}|\psi_{\hat O=\lambda,k}\rangle\langle\psi_{\hat O=\lambda,k}|$

波函数分解：$\psi(\vec r)=\sum_\lambda\psi_\lambda(\vec r)\cdot\langle\psi_\lambda|\psi\rangle$. 例如：$\psi_p(x)=\frac{e^{ipx/\hbar}}{\sqrt{2\pi\hbar}}$. $\langle\psi_p|\psi_{p'}\rangle=\delta(p-p')$

算符在本征基下的矩阵元：$M_{(\lambda,k)(\lambda',k')}=\langle\psi_{\hat O=\lambda,k}|\hat M|\psi_{\hat O=\lambda',k'}\rangle$

动量表象：$\tilde\psi(p)=\langle\psi_p|\psi\rangle=\int\psi_p^*(x)\psi(x)dx$; $\psi(x)=\int\psi_p(x)\tilde\psi(p)dp=\langle\psi_x|\psi\rangle$

测量假设：在ψ量子态下测可观测量$\hat O$. 结果为随机变量λ. λ可取为$\hat O$的所有本征值.

概率 $P(\lambda)=|\langle\psi_{\hat O=\lambda}|\psi\rangle|^2$; 简并情况 $P(\lambda)=\sum_k|\langle\psi_{\hat O=\lambda,k}|\psi\rangle|^2$

坍缩假设：在$|\psi\rangle$态下测$\hat O$，得到结果为λ，则测完后态坍缩到$|\psi_{\hat O=\lambda}\rangle$

简并情况：坍缩到 $A\sum_k|\psi_{\hat O=\lambda,k}\rangle\langle\psi_{\hat O=\lambda,k}|\psi\rangle$ $\quad A$:归一化系数

期待值：$\langle\hat O\rangle=\langle\psi|\hat O|\psi\rangle=\sum_\lambda\lambda\cdot|\langle\psi_{\hat O=\lambda}|\psi\rangle|^2=\sum_\lambda\lambda P(\lambda)$

不确定度：$\sigma_{\hat O}^2=\langle\hat O^2\rangle-\langle\hat O\rangle^2$

不确定度关系：$\sigma_{\hat A}^2\cdot\sigma_{\hat B}^2\geq\frac14|\langle\psi|[\hat A,\hat B]|\psi\rangle|^2$

例：$\sigma_x\cdot\sigma_p\geq\frac{\hbar}{2}$; $\sigma_t\cdot\sigma_E\geq\frac{\hbar}{2}$ 注：$\sigma_t\equiv\frac{\sigma_{\hat Q}}{\left|\frac{d}{dt}\langle\hat Q\rangle\right|}$ ($\hat Q$为某种可观测量)

Schrödinger方程：$i\hbar\frac{\partial}{\partial t}\psi(\vec r,t)=\hat H\psi(\vec r,t)$; $i\hbar\frac{\partial}{\partial t}|\psi(t)\rangle=\hat H|\psi(t)\rangle$; $-i\hbar\frac{\partial}{\partial t}\langle\psi(t)|=\langle\psi(t)|\hat H$

Heisenberg运动方程：$\frac{d}{dt}\langle\hat O\rangle=\langle\frac{\partial\hat O}{\partial t}\rangle+\frac{i}{\hbar}\langle[\hat H,\hat O]\rangle$

Ehrenfest定理：$\frac{d}{dt}\langle\hat x\rangle=\frac{\langle\hat p\rangle}{m}$; $\frac{d}{dt}\langle\hat p\rangle=-\langle\frac{\partial V}{\partial x}\rangle$

定态：$\hat H\psi_E(\vec r)=E\psi_E(\vec r)$; 特解 $\psi_E(\vec r,t)=\psi_E(\vec r)e^{-i\frac{E}{\hbar}t}$

通解 $\psi(\vec r,t)=\text{“}\sum_E\text{”}c_E\psi_E(\vec r)e^{-i\frac{E}{\hbar}t}$ $\quad c_E=\langle\psi_E(\vec r)|\psi(\vec r,t=0)\rangle$

概率流密度：$\vec J(\vec r,t)=\mathrm{Re}\left[\psi^*\frac{\hat{\vec p}}{m}\psi\right]$; $J(x,t)=\frac{i\hbar}{2m}\left(\psi\frac{\partial\psi^*}{\partial x}-\psi^*\frac{\partial\psi}{\partial x}\right)$

连续性方程：$\frac{\partial\rho}{\partial t}+\nabla\cdot\vec J=0$ $\quad\rho(\vec r,t)\equiv|\psi(\vec r,t)|^2$

一维定态问题：$\hat{H}=-\frac{\hbar^2}{2m}\partial_x^2+V(x)$ 求 $\left[-\frac{\hbar^2}{2m}\partial_x^2+V(x)\right]\psi_E(x)=E\psi_E(x)$ 本征值

自由粒子：$\hat{H}=\frac{\hat{p}^2}{2m}$ 定态：$\psi_p(x)=\frac{1}{\sqrt{2\pi\hbar}}e^{ipx/\hbar}$；$E_p=\frac{p^2}{2m}$（对 $p>0$ 二重简并）

通解：$\psi(x,t)=\int dp\,\tilde{\varphi}(p,t=0)\,\psi_p(x)\,e^{-iE_pt/\hbar}$；$\tilde{\varphi}(p,t=0)=\langle\psi_p|\psi(t=0)\rangle=\int\psi_p^*(x)\psi(x,t=0)\,dx$

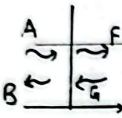
波包：$\psi(x,t=0)=f(x)e^{ip_0x/\hbar}\Rightarrow\psi(x,t)\approx e^{-iE_{p_0}t/\hbar}f(x-V_gt)e^{ip_0x/\hbar}$ 群速度 $V_g=\left.\frac{\partial E_p}{\partial p}\right|_{p=p_0}$

无限深方势阱：$V(x)=\begin{cases}+\infty & \text{else}\\ 0 & a<x<b\end{cases}$ 只有束缚态（忽略隐失波）

本征态 $\psi_n(x)=\begin{cases}0 & ,\text{else}\\ \sqrt{\frac{2}{b-a}}\sin\left(\frac{n\pi}{b-a}(x-a)\right) & ,a<x<b\end{cases}$ $E_n=\frac{\hbar^2}{2m}\left(\frac{n\pi}{b-a}\right)^2$

δ 势：$V(x)=\alpha\delta(x)$ 有散射态和束缚态（对于势阱，只有一个）

边条件：① $\psi|_{x=0^-}=\psi|_{x=0^+}$ $-\frac{\hbar^2}{2m}\left(\left.\frac{\partial\psi}{\partial x}\right|_{x=0^+}-\left.\frac{\partial\psi}{\partial x}\right|_{x=0^-}\right)+\alpha\psi(x=0)=0$



势阱唯一束缚态：$\psi(x)=\sqrt{K}e^{-K|x|}$ $K=-\frac{\alpha m}{\hbar^2}=\frac{\sqrt{-2mE}}{\hbar}$ $E_0=-\frac{m\alpha^2}{2\hbar^2}$

散射态：$\psi_E(x)=\begin{cases}Ae^{ikx}+Be^{-ikx} & x<0\\ Fe^{ikx}+Ge^{-ikx} & x>0\end{cases}$ $k=\frac{\sqrt{2mE}}{\hbar}$ 对 δ 势：$R=\frac{|E_0|}{E+|E_0|}$

反射率 $R=\frac{\text{反射波概率流}}{\text{入射波概率流}}=\frac{|B|^2}{|A|^2}$ 透射率 $T=\frac{\text{透射波概率流}}{\text{入射波概率流}}=\frac{|F|^2}{|A|^2}$ $T=\frac{E}{E+|E_0|}$

一维谐振子：$V(x)=\frac{m}{2}w^2x^2$

升降算符 $\begin{cases}\hat{a}_+=\sqrt{\frac{mw}{2\hbar}}\left(\hat{x}-\frac{i\hat{p}}{mw}\right)\\ \hat{a}_-=\sqrt{\frac{mw}{2\hbar}}\left(\hat{x}+\frac{i\hat{p}}{mw}\right)\end{cases}$ $\begin{cases}\hat{x}=\sqrt{\frac{\hbar}{2mw}}(\hat{a}_++\hat{a}_-)\\ \hat{p}=i\sqrt{\frac{\hbar mw}{2}}(\hat{a}_+-\hat{a}_-)\end{cases}$ $\begin{cases}\hat{a}_+|\psi_n\rangle=\sqrt{n+1}\,|\psi_{n+1}\rangle\\ \hat{a}_-|\psi_n\rangle=\sqrt{n}\,|\psi_{n-1}\rangle\end{cases}$

$[\hat{a}_-,\hat{a}_+]=1$ $\hat{a}_+^\dagger=\hat{a}_-$ $\hat{H}=\hbar w\left(\hat{a}_+\hat{a}_-+\frac{1}{2}\right)$ $[\hat{H},\hat{a}_\pm]=\pm\hbar w\hat{a}_\pm$

本征态：$\psi_0(x)=\left(\frac{mw}{\pi\hbar}\right)^{\frac{1}{4}}e^{-\frac{mw}{2\hbar}x^2}$ $\psi_n(x)=\frac{1}{\sqrt{n!}}(\hat{a}_+^n)\psi_0(x)$ $E_n=\left(n+\frac{1}{2}\right)\hbar w$

期待值时间演化：$\langle\hat{a}_\pm\rangle(t)=e^{\pm iwt}\langle\hat{a}_\pm\rangle(t=0)$ $\langle\hat{a}_{s_1}\cdots\hat{a}_{s_n}\rangle(t)=e^{i(\sum_{i=1}^{n}s_i)wt}\langle\hat{a}_{s_1}\cdots\hat{a}_{s_n}\rangle(t=0)$ $s_i=\pm$

三维定态问题：$\hat{H}=\frac{\hat{\vec{p}}^2}{2m}+V(\hat{\vec{r}})$

直角坐标：$\hat{p}=-i\hbar\nabla$ $[\hat{r}_a,\hat{p}_b]=i\hbar\delta_{ab}$

自由粒子：$\psi_{\vec{p}}(\vec{r})=\frac{1}{(2\pi\hbar)^{3/2}}e^{i\vec{p}\cdot\vec{r}/\hbar}$ $E_{\vec{p}}=\frac{\vec{p}^2}{2m}$

三维各向同性谐振子：$V(\vec{r})=\frac{mw^2}{2}\vec{r}^2$ $\psi_{n_xn_yn_z}(x,y,z)=\psi_{n_x}(x)\psi_{n_y}(y)\psi_{n_z}(z)$ $E_{n_x,n_y,n_z}=\left(n_x+n_y+n_z+\frac{3}{2}\right)\hbar$

球坐标 $\begin{cases}x=r\sin\theta\cos\phi\\ y=r\sin\theta\sin\phi\\ z=r\cos\theta\end{cases}$ 边条件 $\begin{cases}\psi(r,\theta,\phi)=\psi(r,\theta,\phi+2\pi)\\ \lim\limits_{\theta\to0\text{ or }\pi}\psi(r,\theta,\phi)\ \text{与}\ \phi\ \text{无关}\\ \lim\limits_{r\to0}\psi(r,\theta,\phi)\ \text{与}\ \theta,\phi\ \text{无关}\end{cases}$

$$\nabla^2=\frac{1}{r^2}\partial_r(r^2\partial_r)+\frac{1}{r^2\sin\theta}\partial_\theta(\sin\theta\partial_\theta)+\frac{1}{r^2\sin^2\theta}\partial_\phi^2$$

归一化 $\int_0^\infty r^2dr\int_0^\pi\sin\theta\,d\theta\int_0^{2\pi}d\phi\,|\psi(r,\theta,\phi)|^2=1$

中心势问题： $\hat{H} = \frac{\hat{p}^2}{2m} + V(r)$

定态本征解： $\psi_{nlm}(r,\theta,\phi) = R_{n,l}(r)\, Y_l^m(\theta,\phi)$ $\quad E_{nlm} = E_{nl}$ $\quad (l=0,1,2,\cdots \quad m=-l,\cdots,l)$

球谐函数 $Y_l^m(\theta,\phi) = \epsilon\sqrt{\frac{2l+1}{4\pi}\frac{(l-|m|)!}{(l+|m|)!}}\, P_l^m(\cos\theta)\, e^{im\phi}$ $\quad \epsilon = \begin{cases} 1 & m\geq 0 \\ (-)^m & m<0 \end{cases}$

连带Legendre函数 $P_l^m(x) = (1-x^2)^{|m|/2}\left(\frac{d}{dx}\right)^{|m|} P_l(x)$

Legendre多项式： $P_l(x) = \frac{1}{2^l l!}\left(\frac{d}{dx}\right)^l \left[(x^2-1)^l\right]$

重要性质 $\left[\frac{1}{\sin\theta}\partial_\theta(\sin\theta\,\partial_\theta) + \frac{1}{\sin^2\theta}\partial_\phi^2\right] Y_l^m = -l(l+1)\, Y_l^m$

$\partial_\phi Y_l^m = im\, Y_l^m$

归一化： $\int_0^\pi \sin\theta\, d\theta \int_0^{2\pi} Y_l^{m*}\cdot Y_{l'}^{m'}\, d\phi = \delta_{ll'}\delta_{mm'}$

径向方程： $\left[-\frac{\hbar^2}{2m}\frac{1}{r^2}\partial_r(r^2\partial_r) + \frac{\hbar^2}{2m}\frac{l(l+1)}{r^2} + V(r)\right] R(r) = E\cdot R(r)$

$u(r)=R(r)\cdot r$ 变形： $\left[-\frac{\hbar^2}{2m}\partial_r^2 + \frac{\hbar^2}{2m}\frac{l(l+1)}{r^2} + V(r)\right] u(r) = E\cdot u(r)$

归一化： $\int_0^\infty |R(r)|^2 r^2\, dr = \int_0^\infty |u(r)|^2 dr = 1$

渐近行为： $r\to 0$ 时 $R(r)\sim r^l \quad u(r)\sim r^{l+1}$

自由粒子： $R_{kl}(r) = A\, j_l(kr) \qquad E_{klm} = \frac{\hbar^2 k^2}{2m}$

球Bessel函数： $j_l(x) = (-x)^l\left(\frac{1}{x}\frac{d}{dx}\right)^l\left(\frac{\sin x}{x}\right)$ $\quad x\to 0$ 时 $\sim \frac{x^l}{(2l+1)!!}$ $\quad x\to\infty$ 时 $\sim \frac{\sin(x-\frac{l}{2}\pi)}{x}$

球Neumann函数 $n_l(x) = -(-x)^l\left(\frac{1}{x}\frac{d}{dx}\right)^l\left(\frac{\cos x}{x}\right)$ $\quad x\to 0$ 时 $\sim -(2l-1)!!\, x^{-l-1}$ $\quad x\to\infty$ 时 $\sim -\frac{\cos(x-\frac{l}{2}\pi)}{x}$

一维谐振子： $\psi_n(x) \propto H_n(\xi)\, e^{-\frac{\xi^2}{2}}$ $\quad \xi = \sqrt{\frac{m\omega}{\hbar}}\, x$ $\quad H_n(\xi) = \sum_{j=0}^{n} C_j \xi^j$

$C_{j+2} = C_j \frac{2j+1-K}{(j+1)(j+2)}$ $\quad K = \frac{2E}{\hbar\omega}$ $\quad E_n = \hbar\omega(n+\frac{1}{2})$

三维谐振子： $\psi_{nlm}(r,\theta,\phi) = R_{nl}(r)\, Y_l^m(\theta,\phi)$ $\quad R_{nl}(r) = f_{nl}(\xi)\,\xi^l e^{-\frac{\xi^2}{2}}$ $\quad f_{nl}(\xi) = \sum_{j=0}^{n-l} c_j \xi^j$ $\quad \xi = \sqrt{\frac{m\omega}{\hbar}}\, r$

$c_{j+2} = c_j \frac{2j+2l+3-K}{(j+2)(j+2l+3)}$ $\quad c_0\neq 0$ $\quad K = \frac{2E}{\hbar\omega}$ $\quad E_n = \hbar\omega(n+\frac{3}{2})$

氢原子 $(V=-\frac{A}{r})$： $\psi_{nlm}(r,\theta,\phi) = R_{nl}(r)\, Y_l^m(\theta,\phi)$ $\quad R_{nl}(r) = V_{nl}(\rho)\,\rho^l e^{-\rho}$ $\quad V_{nl}(\rho) = \sum_{j=0}^{n-l-1} c_j \rho^j$ $\quad \rho = r\cdot\frac{\sqrt{-2mE}}{\hbar}$

$c_{j+1} = c_j\cdot\frac{2(j+l+1)-\rho_0}{(j+1)(j+2l+2)}$ $\quad c_0\neq 0$ $\quad \rho_0 = \frac{2mA}{\hbar\sqrt{-2mE}}$; $E_{nl} = \frac{E_1}{n^2}$; $E_1 = -\frac{mA^2}{2\hbar^2}$ $\quad A = \frac{e^2}{4\pi\varepsilon_0}$

基态 $\psi_{100}(r,\theta,\phi) = \frac{1}{\sqrt{\pi a^3}}\, e^{-\frac{r}{a}}$ $\quad$ Bohr半径 $a = \frac{\hbar^2}{mA}$ $\quad E_1 = -\frac{\hbar^2}{2ma^2}$

轨道角动量： $\hat{\vec{L}} = \hat{\vec{r}}\times\hat{\vec{p}}$ $\quad \hat{L}_a = \epsilon_{abc}\hat{r}_b\hat{p}_c$

对易关系：与矢量算符 $\vec{V}$ (如 $\vec{r},\vec{p},\vec{L}$) ： $[\hat{L}_a, \hat{V}_b] = i\hbar\,\epsilon_{abc}\hat{V}_c$

与标量算符 $\hat{A}$ (如 V^2) ： $[\hat{L}_a, \hat{A}] = 0$

球坐标下本征函数 $|l,m\rangle$ 对应 $R(r)Y_l^m(\theta,\phi)$ 其中 $R(r)$ 任意

$\hat{L}_z|l,m\rangle = m\hbar|l,m\rangle$

$\hat{L}^2|l,m\rangle = l(l+1)\hbar^2|l,m\rangle$

升降算符 $\begin{cases} \hat{L}_+ = \hat{L}_x + i\hat{L}_y \\ \hat{L}_- = \hat{L}_x - i\hat{L}_y \end{cases}$ $\begin{cases} \hat{L}_+|l,m\rangle = \hbar\sqrt{(l-m)(l+m+1)}\,|l,m+1\rangle \\ \hat{L}_-|l,m\rangle = \hbar\sqrt{(l+m)(l-m+1)}\,|l,m-1\rangle \end{cases}$

$[\hat{L}_z, \hat{L}_\pm] = \pm i\hbar \hat{L}_\pm$ $[\hat{L}^2, \hat{L}_\pm] = 0$ $\hat{L}_\pm^\dagger = \hat{L}_\mp$ $[\hat{L}_+, \hat{L}_-] = 2\hbar \hat{L}_z$

$\hat{L}^2 = \hat{L}_+\hat{L}_- + \hat{L}_z(\hat{L}_z - \hbar) = \hat{L}_-\hat{L}_+ + \hat{L}_z(\hat{L}_z + \hbar)$

球坐标下：$\hat{L}_z = -i\hbar\frac{\partial}{\partial\phi}$ $\hat{L}_\pm = \pm\hbar e^{\pm i\phi}\left(\frac{\partial}{\partial\theta} \pm i\cot\theta\frac{\partial}{\partial\phi}\right)$

自旋：本征基 $|s, s_z\rangle$ $s_z = -s, \cdots, +s$

自旋角动量算符：$[\hat{S}_a, \hat{S}_b] = i\hbar\epsilon_{abc}\hat{S}_c$ $\hat{S}_z|s,s_z\rangle = s_z\hbar|s,s_z\rangle$

升降算符 $\begin{cases} \hat{S}_+ = \hat{S}_x + i\hat{S}_y \\ \hat{S}_- = \hat{S}_x - i\hat{S}_y \end{cases}$ $\begin{cases} \hat{S}_+|s,s_z\rangle = \hbar\sqrt{(s-s_z)(s+s_z+1)}\,|s,s_z+1\rangle \\ \hat{S}_-|s,s_z\rangle = \hbar\sqrt{(s+s_z)(s-s_z+1)}\,|s,s_z-1\rangle \end{cases}$

自旋$\frac{1}{2}$：本征基 $|\uparrow\rangle = |\frac{1}{2},\frac{1}{2}\rangle$ $|\downarrow\rangle = |\frac{1}{2},-\frac{1}{2}\rangle$

$\hat{S}_a = \frac{\hbar}{2}\sigma_a$ 泡利矩阵：$\sigma_x = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ $\sigma_y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$ $\sigma_z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ $\sigma_0 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

性质：$\sigma_i^\dagger = \sigma_i$ $\sigma_a\sigma_b = i\epsilon_{abc}\sigma_c + \delta_{ab}\sigma_0$ $[\sigma_a, \sigma_b] = 2i\epsilon_{abc}\sigma_c$ $\{\sigma_a, \sigma_b\} = 2\delta_{ab}\sigma_0$

矩阵 $\overleftrightarrow{M} = \vec{u}\cdot\vec{\sigma} = \sum_{a=x,y,z} u_a\sigma_a$ 本征值为$\pm|\vec{u}|$；$u_0\sigma_0 + \vec{u}\cdot\vec{\sigma}$ 本征值为 $u_0 \pm |\vec{u}|$

一般自旋$\frac{1}{2}$态：$|\psi\rangle = \psi_\uparrow|\uparrow\rangle + \psi_\downarrow|\downarrow\rangle$ 写作$\begin{bmatrix} \psi_\uparrow \\ \psi_\downarrow \end{bmatrix}$ 归一化 $|\psi_\uparrow|^2 + |\psi_\downarrow|^2 = 1$

旋量波函数 $|\psi\rangle = \psi_\uparrow(\vec{r})|\uparrow\rangle + \psi_\downarrow(\vec{r})|\downarrow\rangle$ 写作$\begin{bmatrix} \psi_\uparrow(\vec{r}) \\ \psi_\downarrow(\vec{r}) \end{bmatrix}$ 归一化 $\int d^3\vec{r}\,(|\psi_\uparrow(\vec{r})|^2 + |\psi_\downarrow(\vec{r})|^2) = 1$

全同粒子：交换对称性假设。N個粒子经典状态与任意置换σ满足

$$\psi(\vec{r}_1, \cdots, \vec{r}_n) = \begin{cases} \psi(\vec{r}_{\sigma(1)}, \cdots, \vec{r}_{\sigma(n)}) & \forall\sigma \text{ 玻色子} \\ \mathrm{sgn}(\sigma)\,\psi(\vec{r}_{\sigma(1)}, \cdots, \vec{r}_{\sigma(n)}) & \forall\sigma \text{ 费米子} \end{cases}$$

归一化：$\int d\vec{r}_1 \cdots d\vec{r}_n\,|\psi|^2 = 1$

两个无相互作用的全同粒子：$\hat{H} = \hat{H}_1(\hat{r}_1, \hat{p}_1) + \hat{H}_2(\hat{r}_2, \hat{p}_2)$

玻色子本征基 $\psi^{(B)}_{k_1k_2}(\vec{r}_1, \vec{r}_2) = \begin{cases} \psi_k(\vec{r}_1)\psi_k(\vec{r}_2) & k_1 = k_2 = k \\ \frac{1}{\sqrt{2}}[\psi_{k_1}(\vec{r}_1)\psi_{k_2}(\vec{r}_2) + \psi_{k_2}(\vec{r}_1)\psi_{k_1}(\vec{r}_2)] & k_1 < k_2 \end{cases}$

费米子本征基 $\psi^{(F)}_{k_1,k_2}(\vec{r}_1, \vec{r}_2) = \frac{1}{\sqrt{2}}[\psi_{k_1}(\vec{r}_1)\psi_{k_2}(\vec{r}_2) - \psi_{k_2}(\vec{r}_1)\psi_{k_1}(\vec{r}_2)]$ $k_1 < k_2$

总能量 $E^{(B,F)}_{k_1,k_2} = E_{k_1} + E_{k_2}$

归一化：$\iint |\psi^{(B,F)}_{k_1,k_2}(\vec{r}_1, \vec{r}_2)|^2\,d\vec{r}_1\,d\vec{r}_2 = 1$

单体算符性质：$\langle\psi^{(B,F)}_{nl}|\,\hat{O}(\vec{r}_1, \vec{p}_1) + \hat{O}(\vec{r}_2, \vec{p}_2)\,|\psi^{(B,F)}_{nl}\rangle = \langle\psi_n|\hat{O}|\psi_n\rangle + \langle\psi_l|\hat{O}|\psi_l\rangle$